

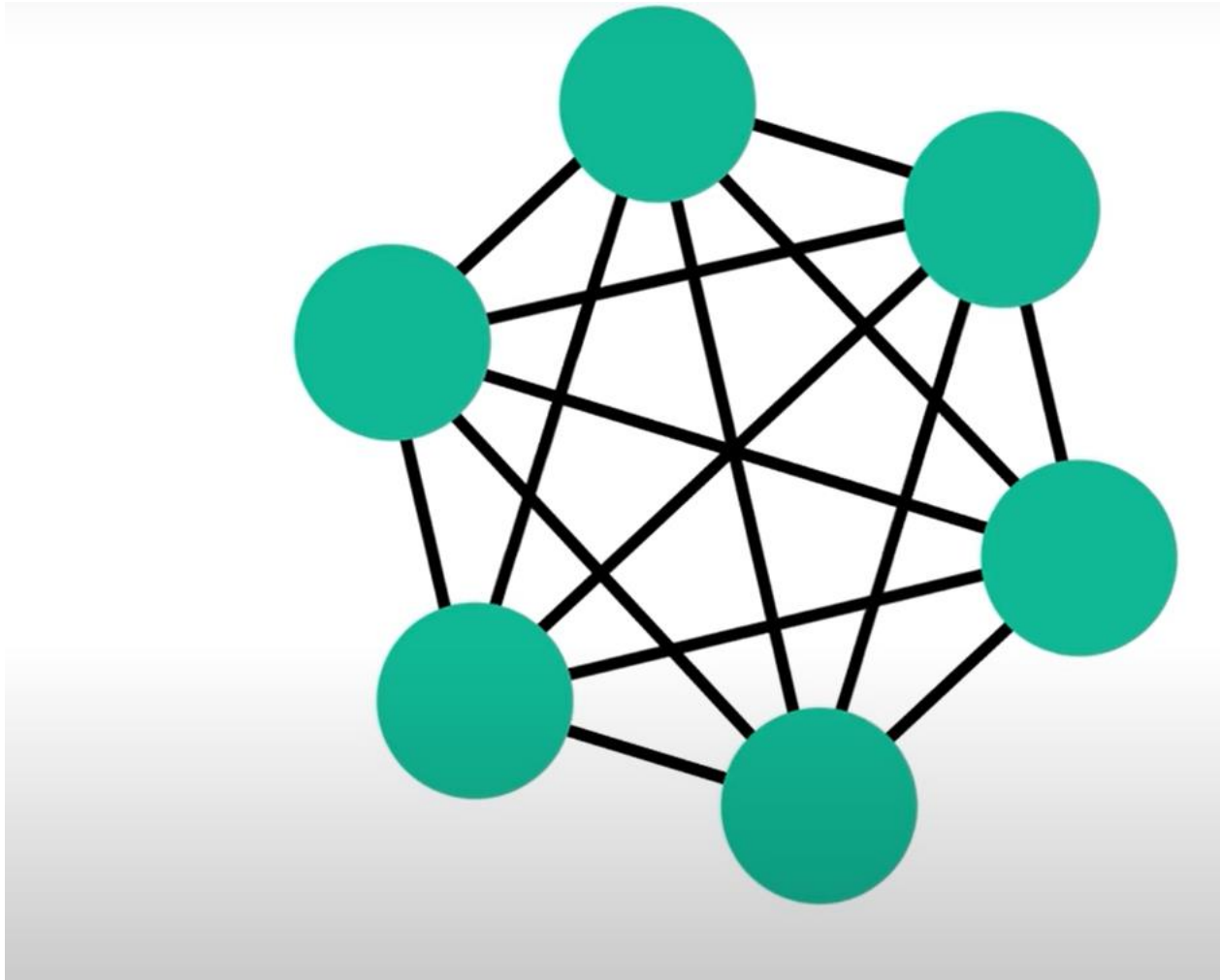
ZigBee
Anusha N

Zigbee

- ZigBee is the most popular industry wireless mesh networking standard for connecting sensors, instrumentation and control systems.
- ZigBee, a specification for communication in a wireless personal area network (WPAN), has been called the "Internet of things."

TECHNOLOGY	DATA RATE
	$\geq 1\text{Gbps}$
	1-3 Mbps
	250 Kbps



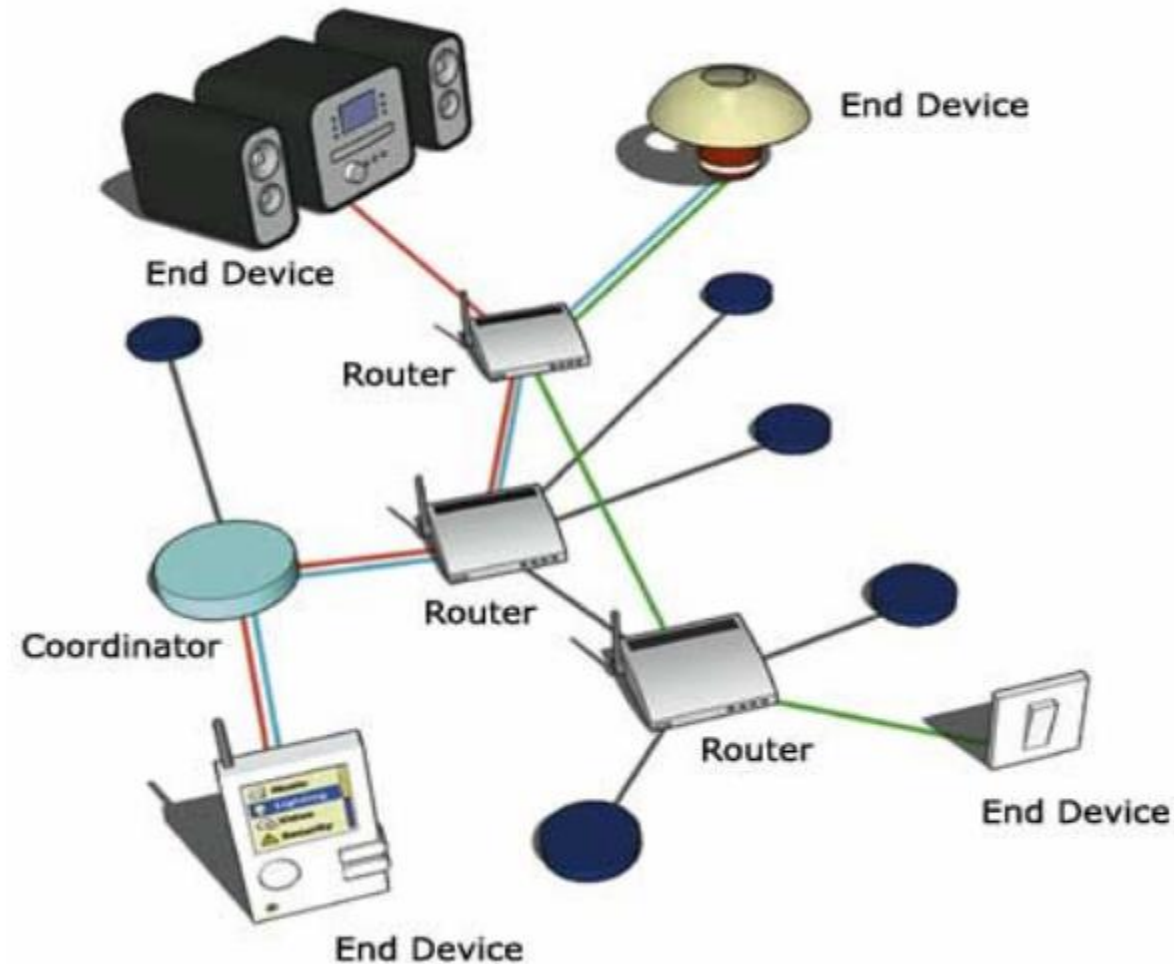


Zigbee

ZigBee applications include:

- Home and office automation
- Industrial automation
- Medical monitoring
- Low-power sensors

- ZigBee targets the application domain of low power, low duty cycle and low data rate requirement devices.



INTRODUCTION



- Technological standard created for control and sensor networks.
- Based on the IEEE 802.15.4 standard. it can provide the wireless personal area network.
- ZigBee specification is to be simpler and less expensive compare with the other WPN's such as Bluetooth.
- Designed for low power consumption allowing batteries to essentially last for ever.

➤A wireless technological device which is popular for extremely low power, and low bit rate wireless PAN technology called “**zigBee**”.

➤Designed for

- wireless automation
- lower data tasks

➤ZigBee enables new opportunities for wireless sensors and control networks, zigBee is standard based

➤ Used globally, reliable and self healing, support large of nodes, easy to deploy ,very long battery life and secure .

- ZigBee is focused on control and automation.
- ZigBee uses low data rates, low cost, low power consumption and work with small packet device.
- ZigBee network can support a larger number of devices and a longer range between devices than Bluetooth.



Comparison of Wireless Standards			
Wireless Parameter	Bluetooth	Wi-Fi	ZigBee
Frequency band	2.4 GHz	2.4 GHz	2.4 GHz
Physical/MAC layers	IEEE 802.15.1	IEEE 802.11b	IEEE 802.15.4
Range	9 m	75 to 90 m	Indoors: up to 30 m Outdoors (line of sight): up to 100 m
Current consumption	60 mA (Tx mode)	400 mA (Tx mode) 20 mA (Standby mode)	25-35 mA (Tx mode) 3 μ A (Standby mode)
Raw data rate	1 Mbps	11 Mbps	250 Kbps
Protocol stack size	250 KB	1 MB	32 KB 4 KB (for limited function end devices)
Typical network join time	>3 sec	variable, 1 sec typically	30 ms typically
Interference avoidance method	FHSS (frequency-hopping spread spectrum)	DSSS (direct-sequence spread spectrum)	DSSS (direct-sequence spread spectrum)
Minimum quiet bandwidth required	15 MHz (dynamic)	22 MHz (static)	3 MHz (static)
Maximum number of nodes per network	7	32 per access point	64 K
Number of channels	19	13	16

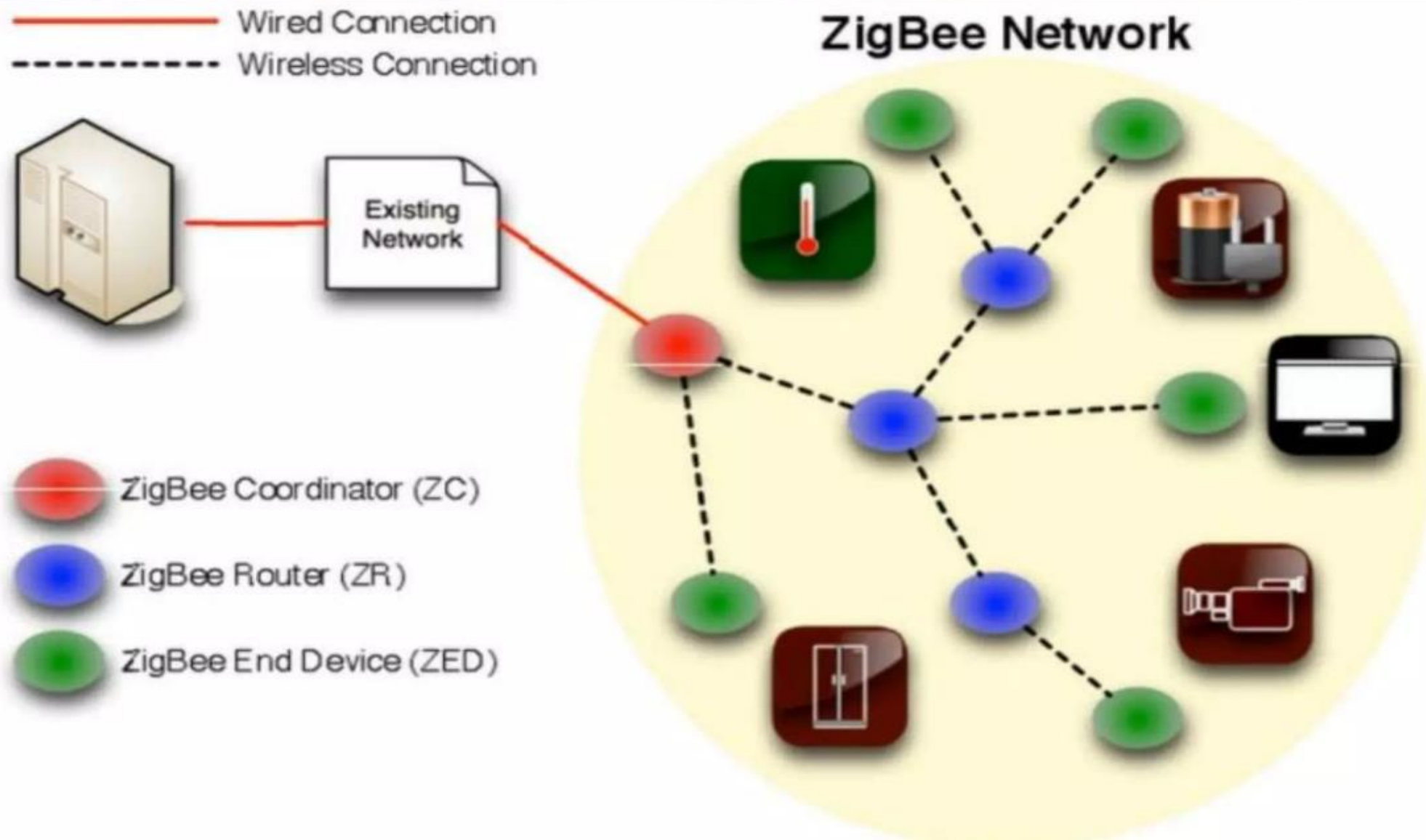
Zigbee

- Low power consumption, simply implemented
- Users expect batteries to last many months to years
- Bluetooth has many different modes and states depending upon your latency and power requirements such as sniff, park, hold, active, etc.; ZigBee/IEEE 802.15.4 has active (transmit/receive) or sleep
- Even mains powered equipment needs to be conscious of energy. ZigBee devices will be more ecological than its predecessors saving megawatts at it full deployment.

Basics of ZigBee

- ZigBee is developed by ZigBee Alliance, ZigBee is a competitor of 6LOWPAN.
- ZigBee uses IEEE 802.15.4 at MAC and Physical Layer.
- ZigBee is designed for low-cost and low-power wireless IoT Networks.
- ZigBee is used in low data rate applications that requires long battery life and secure networking.
- ZigBee is simpler and less expensive than other WPANs such as Bluetooth and Wi-Fi.
- ZigBee operates at short range around 10m to 20m and using mesh networking range can be extended up to 500m.

ARCHITECTURE





ZIGBEE DEVICE TYPES

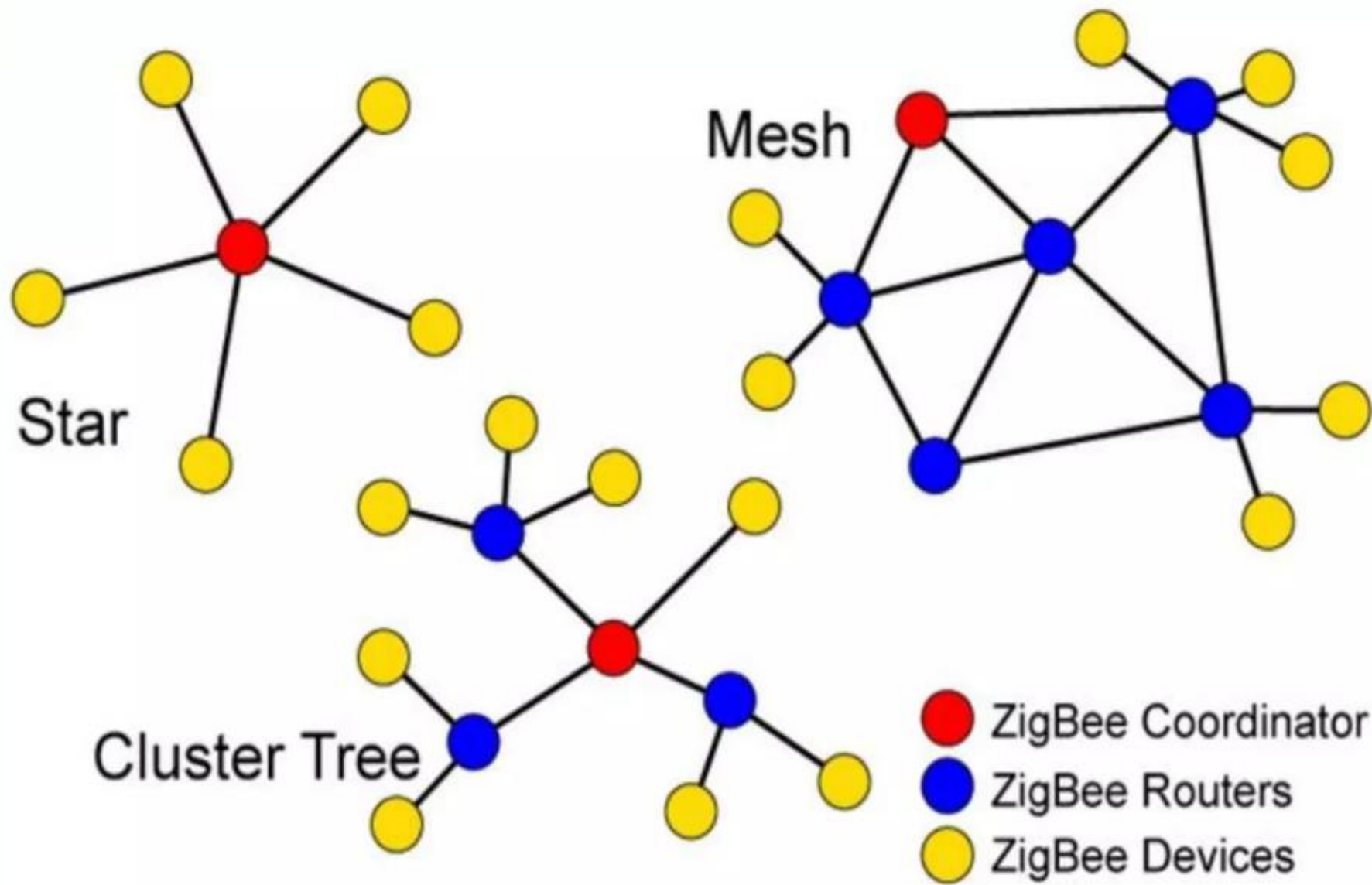
zigbee system structure consists of three different types of devices such as

- Zigbee Coordinator(ZC)
- Zigbee Router(ZR)
- Zigbee End devices (ZED)

NETWORK TOPOLOGIES

The number of routers ,coordinators and end devices depends on the type of network such as

- Star Topology
- Mesh Topology
- Cluster tree Topology



Zigbee Topologies



Zigbee Devices

Zigbee Coordinator

- In Zigbee network it has one coordinator that can store information
- This node can control a complete network, that can communicate with an external network
- ZCs and ZRs require high power, so it can be battery powered

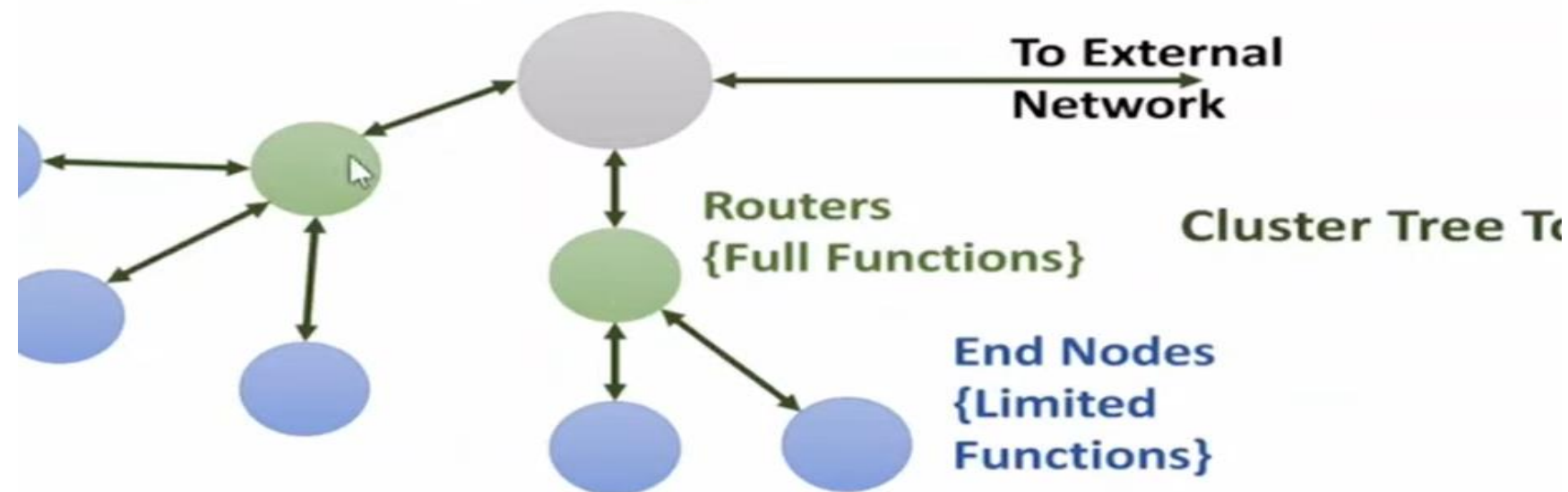
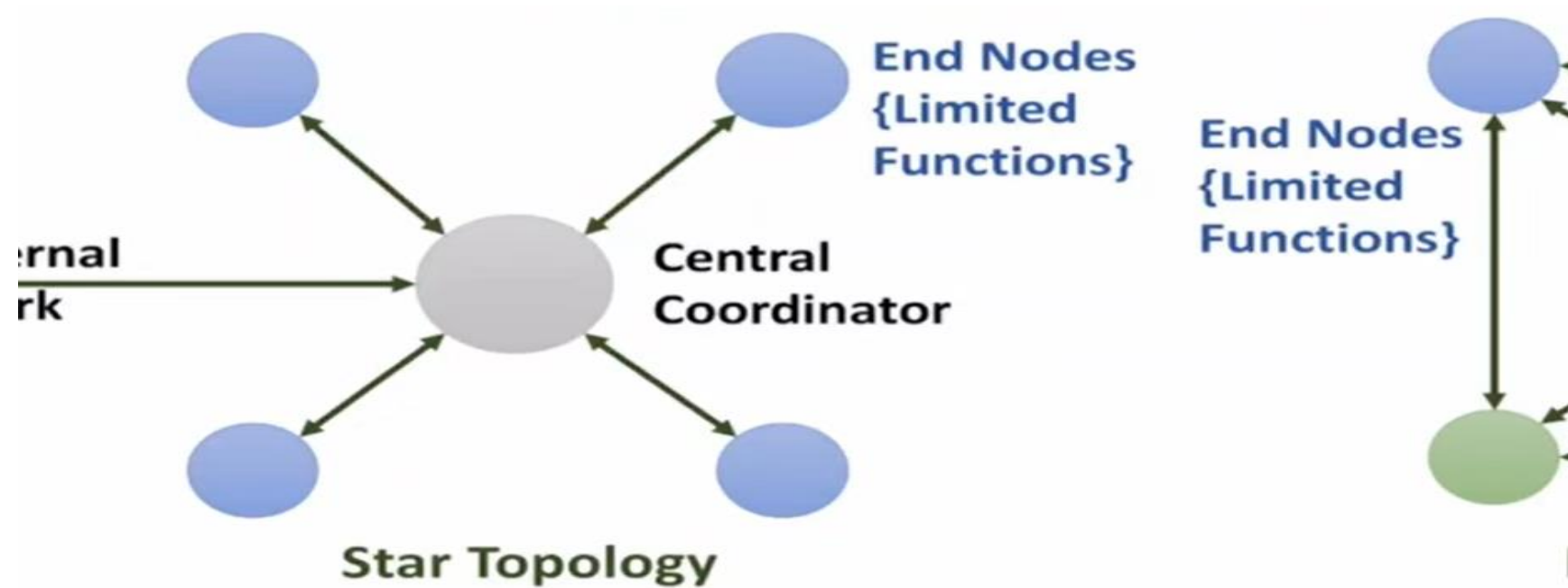
Zigbee Router(Full function device)

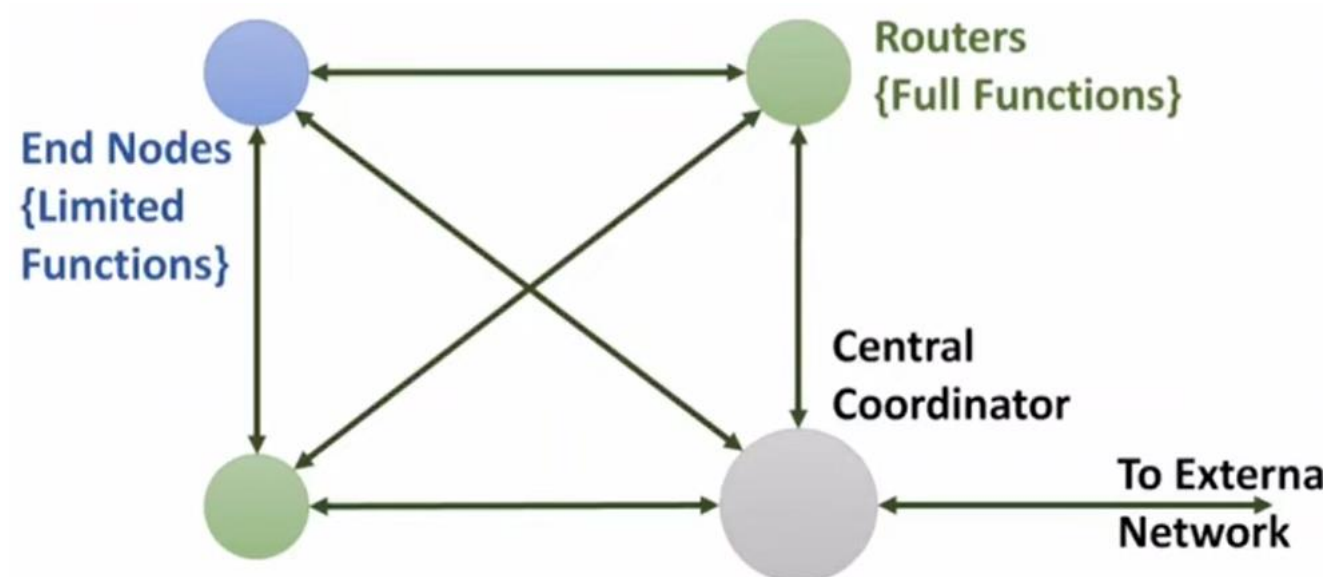
- In Zigbee network, it can route the data with interconnected nodes
- It needs less memory than ZCs
- It can also a coordinator in the network

Zigbee Devices

Zigbee End Device(Reduced Function Device)

- It is the least expensive node with minimum
- It has a minimum power requirement
- It can only communicate with connected ZR or ZC
- It communicates periodically for effective battery utilization





Protocol Stack of ZigBee

Application Layer	ZigBee
Network Layer (Routing/ Network)	ZigBee
Data Link Layer (MAC – CSMA/CA)	IEEE 802.15.4
Physical Layer (2.4GHz, 250kbps)	IEEE 802.15.4

Protocol Stack of ZigBee

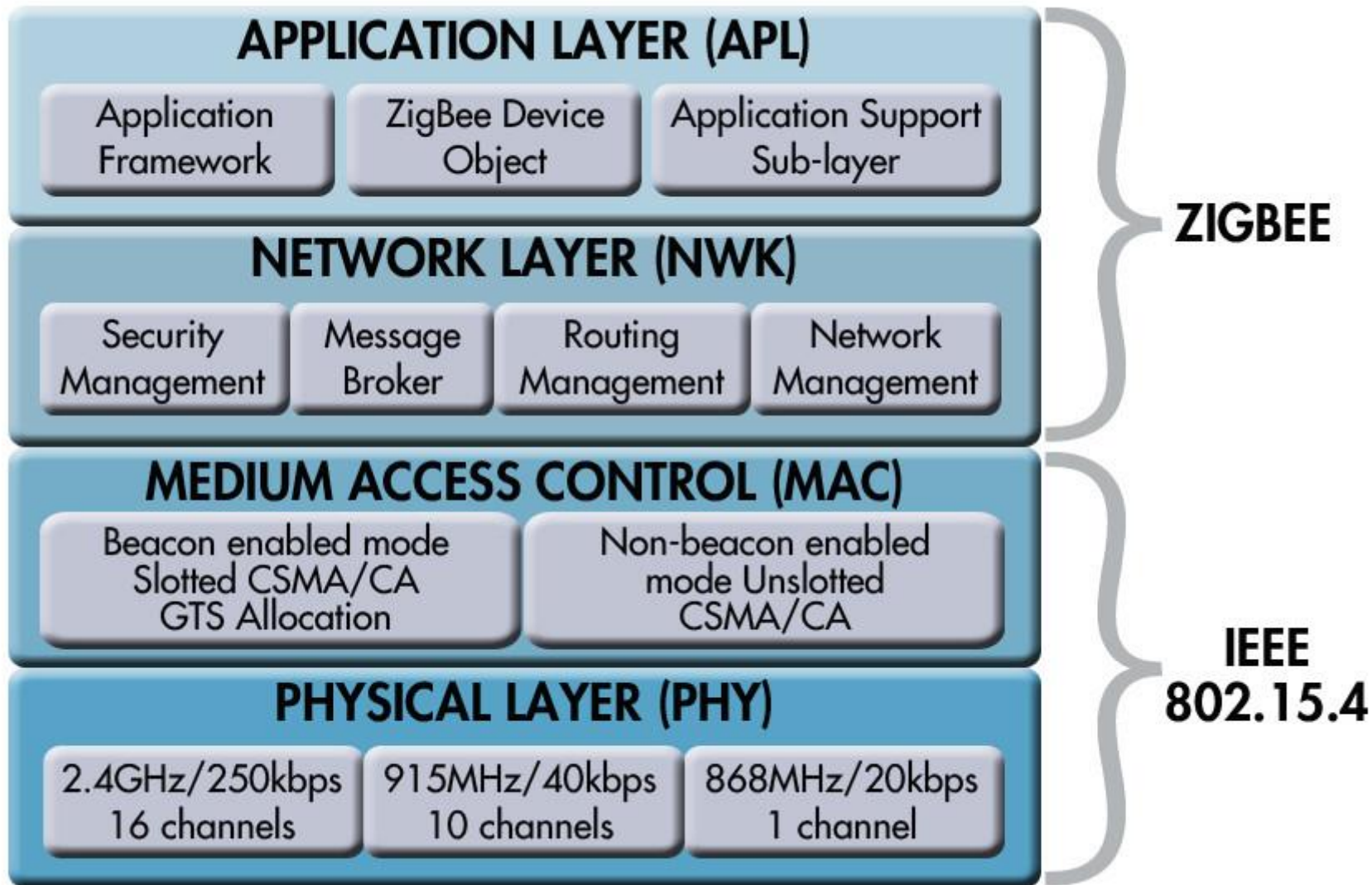
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- ZigBee supports different network topologies like:
 - STAR
 - Peer to Peer (Mesh)
 - Cluster Tree
- ZigBee can be specified as three different devices:
 - ZigBee Coordinator
 - ZigBee Router (Full Function Device)
 - ZigBee End Device (Reduced Function Device)

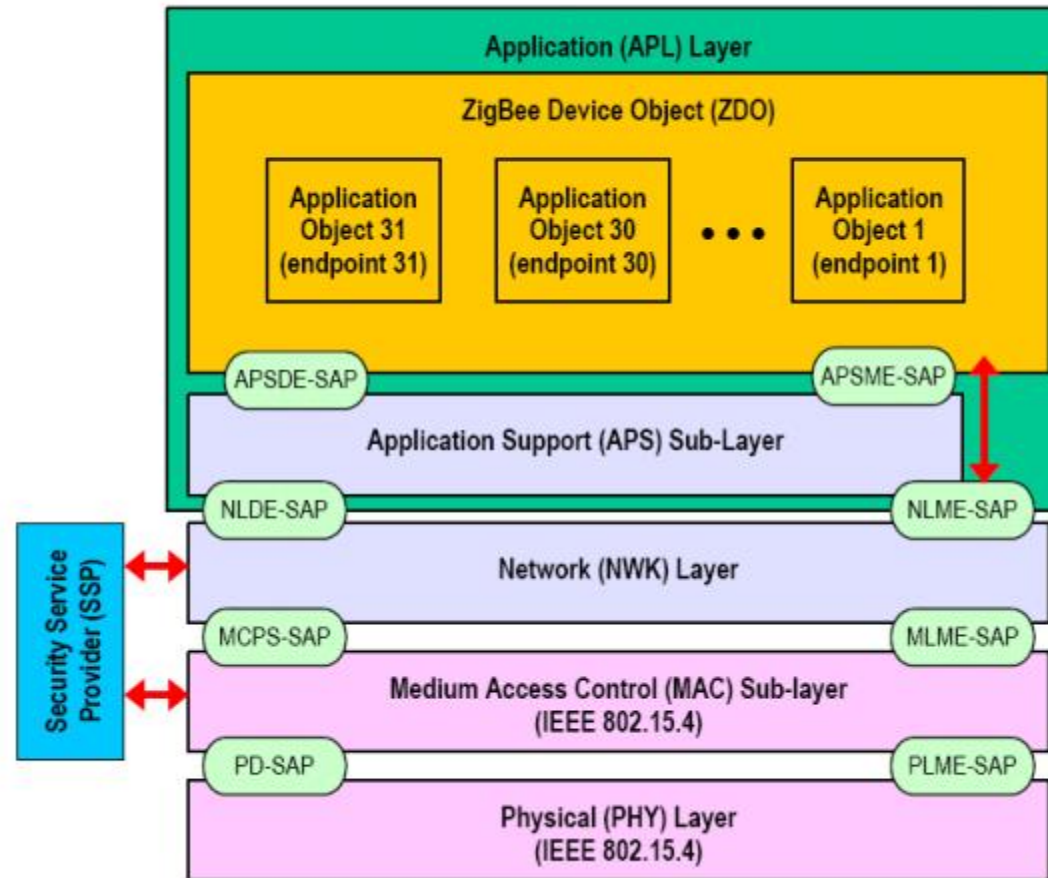
Protocol Stack of IEEE 802.15.4

Application Layer	ZigBee, 6LOWPAN
Network Layer (Routing/ Network)	ZigBee, 6LOWPAN
Data Link Layer (MAC – CSMA/CA)	IEEE 802.15.4
Physical Layer (2.4GHz, 250kbps)	IEEE 802.15.4

- Media Access Control (MAC) is done by CSMA/CA (Carrier Sense Multiple Access Collision Avoidance).
- MAC defines medium access and flow control mechanisms.
- The Physical Layer defines operational frequency, transmission power, and modulation schemes.
- IEEE 802.15.4 utilizes a Direct Sequence Spread Spectrum (DSSS) coding scheme to transmit information.



ZigBee Stack Layers



Basics of LR-WPAN

- LR-WPAN – Low-Rate Wireless Personal Area Network
- It is IEEE 802.15.4 standard.
- Generally, It is operated at a 2.4 GHz ISM band. {868 MHz available in Europe and 915 MHz available in US}
- IEEE 802.15.4 is used for low-power wireless connectivity solutions like ZigBee, 6LOWPAN, and many more.

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- IEEE 802.15.4 is used for low-power wireless connectivity solutions like ZigBee, 6LOWPAN, and many more.
- It is operated at a low data rate with good performance of battery life.
- It is operated with lower distance communication {<100m}
- IEEE 802.15.4 defines the characteristics of the physical layer and MAC.



ZigBee Architecture

- ZigBee and 802.15.4
 - ZigBee sits on top of 802.15.4 physical (PHY) and medium-access control (MAC) layers
 - ZigBee uses only the 2003 version of 802.15.4
 - All existing ZigBee commercial devices use 2.4 GHz S-Band as 2003 version of 802.15.4 does not allow sufficient bandwidth on other frequencies
 - 2006 version adds improved data-transfer rates for 868 and 900 MHz
 - 802.15.4 offers 16 channels on the 2.4 GHz, numbered 11 to 26.

ZigBee Architecture

- Application Support Sublayer (APS) Layer:
 - Multiplexing/demultiplexing: It forwards the network layer messages to the appropriate application objects, according to their endpoint ID (each application is allocated an endpoint ID)
 - Binding: Maintains local binding table i.e. records remote nodes & endpoints which have registered to receive messages from local endpoint
 - 64-bit IEEE to 16-bit ZigBee network node address mapping
 - Management of end to end acknowledgements
 - Fragmentation, Group addressing, Security_ 

ZigBee Architecture

- ZigBee Device Object (ZDO) Layer:
 - Specific application running on endpoint 0
 - Designed to manage the state of the ZigBee node
 - ZDO application implements the interfaces defined by ZigBee device profile (ZDP)
 - These primitives encapsulate the 802.15.4 network formation primitives of the ZigBee network layer as well as additional primitives supporting the concept of binding.

Operating Frequency Bands (Only one channel will be selected for use in a network):

- 1.Channel 0:** 868 MHz (Europe)
- 2.Channel 1-10:** 915 MHz (the US and Australia)
- 3.Channel 11-26:** 2.4 GHz (Across the World)

Features of Zigbee:

- 1. Stochastic addressing:** A device is assigned a random address and announced. Mechanism for address conflict resolution. Parents node don't need to maintain assigned address table.
- 2. Link Management:** Each node maintains quality of links to neighbors. Link quality is used as link cost in routing.
- 3. Frequency Agility:** Nodes experience interference report to channel manager, which then selects another channel
- 4. Asymmetric Link:** Each node has different transmit power and sensitivity. Paths may be asymmetric.
- 5. Power Management:** Routers and Coordinators use main power. End Devices use batteries.

Network (NWK) Layer

The network layer ensures the proper operation of the underlying MAC layer and provides an interface to the application layer.

The network layer supports star, tree and mesh topologies.

Among other things, this is the layer where networks are started, joined, left and discovered.

A node can join the network either directly or through association.

To join directly, the system designer must somehow add a node's extended address into the neighbour table of a device. The direct joining device will issue an orphan scan, and the node with the matching extended address (in its neighbour table) will respond, allowing the device to join.

To join by association, a node sends out a beacon request on a channel, repeating the beacon request on other channels until it finds an acceptable network to join.

Application (APL) Layer

The APL layer is made up of several sublayers.

The ovals symbolize the interface, called service access points (SAP), between different sublayer entities.

Application Support Sublayer (APS)

The APS sublayer is responsible for:

- binding tables
- message forwarding between bound devices
- group address definition and management
- address mapping from 64-bit extended addresses to 16-bit NWK addresses
- fragmentation and reassembly of packets • reliable data transport

Application Framework

The application framework is an execution environment for application objects to send and receive data

An application object actually implements the application; it can be a light bulb, a light switch, an LED, an I/O line, etc. The application profile is run by the application objects.

Each application object is addressed through its corresponding endpoint. Endpoint numbers range from 1 to 240. Endpoint 0 is the address of the ZigBee Device Object (ZDO). Endpoint 255 is the broadcast address, i.e., messages are sent to all of the endpoints on a particular node. Endpoints 241 through 254 are reserved for future use.

ZigBee Device Object (ZDO)

specifically it is responsible for:

- initializing the APS sublayer and the NWK layer
- defining the operating mode of the device (i.e., coordinator, router, or end device)
- device discovery and determination of which application services the device provides
- initiating and/or responding to binding requests
- security management

ZigBee Addressing

Before joining a ZigBee network, a device with an IEEE 802.15 compliant radio has a 64-bit address.

When the device joins a Zigbee network, it receives a 16-bit address called the NWK address. Either of these addresses, the 64-bit extended address or the NWK address, can be used within the PAN to communicate with a device. The coordinator of a ZigBee network always has a NWK address of "0.

ZigBee Messaging

After a device has joined the ZigBee network, it can send commands to other devices on the same network. There are two ways to address a device within the ZigBee network: direct addressing and indirect addressing.

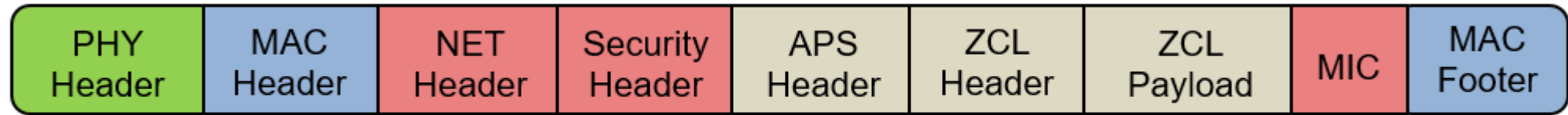
Direct addressing requires the sending device to know three kinds of information regarding the receiving device:
1. Address 2. Endpoint Number 3. Cluster ID

Indirect addressing requires that the above three types of information be committed to a binding table. The sending device only needs to know its own address, endpoint number and cluster ID.

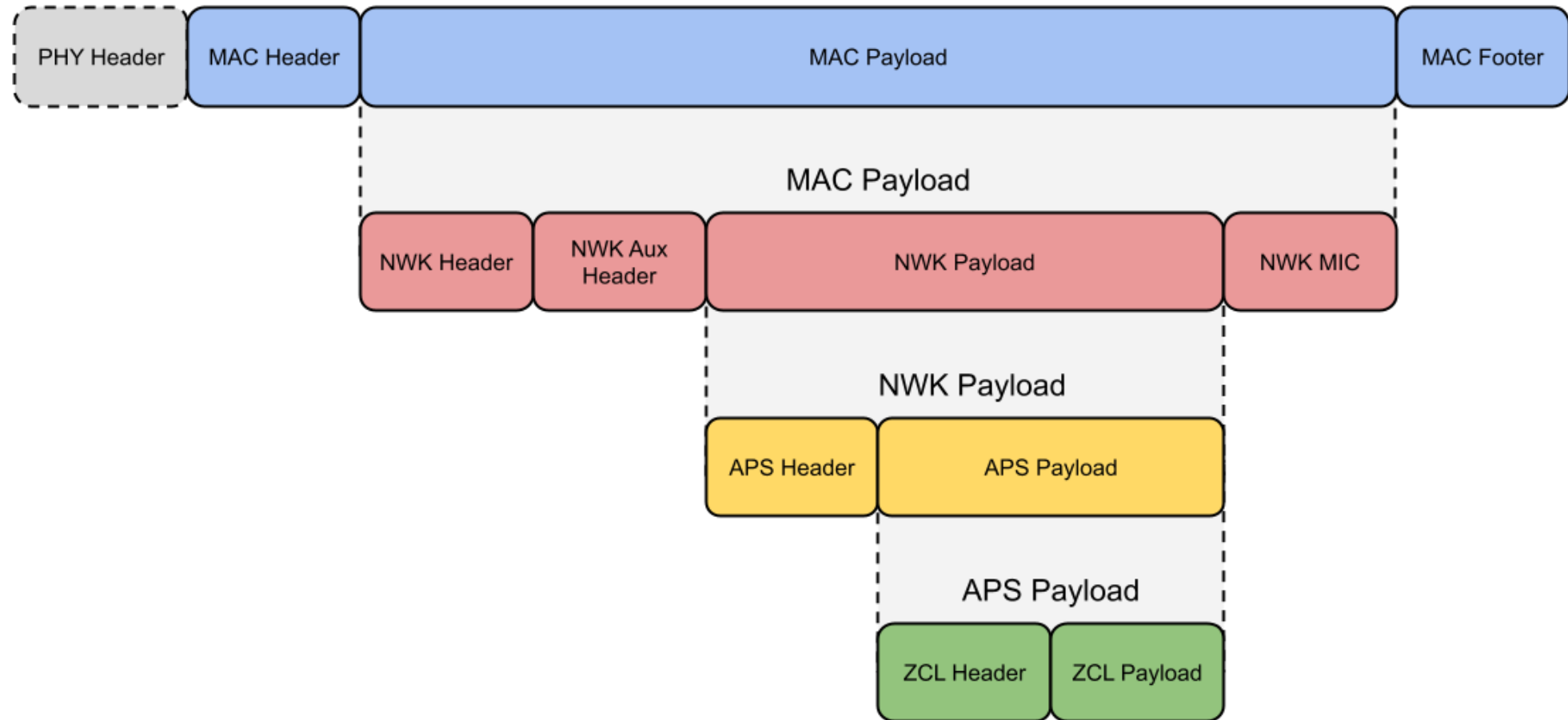
Broadcast Addressing

There are two distinct levels of broadcast addresses used in a ZigBee network. One is a broadcast packet with a MAC layer destination address of 0xFFFF.

The other broadcast address is the use of endpoint number 0xFF to send a message to all of the endpoints on the specified device.



Format of IEEE 802.15.4 / ZigBee Frame
ZigBee frame



Group Addressing

An application can assign multiple devices and specific endpoints on those devices to a single group address. The source node would need to provide the cluster ID, profile ID and source endpoint.

Zigbee Frame Structure

A beacon frame, used by a coordinator to transmit beacons.

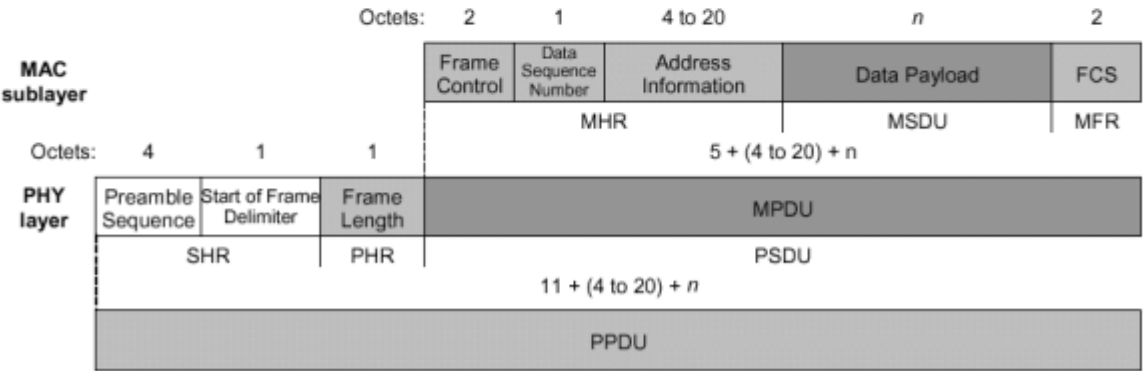
A data frame, used for all transfers of data.

An acknowledgment frame, used for confirming successful frame reception.

A MAC command frame, used for handling all MAC peer entity control transfers.

The data frame is illustrated below:

Preamble Sequence	4 Octets
Start of Frame Delimiter	1 Octet
Frame Length	1 Octet
Frame Control	2 Octets
Data Sequence Number	1 Octet
Address Information	4 – 20 Octets
Frame Check Sequence	2 Octets



Super Frame Structure

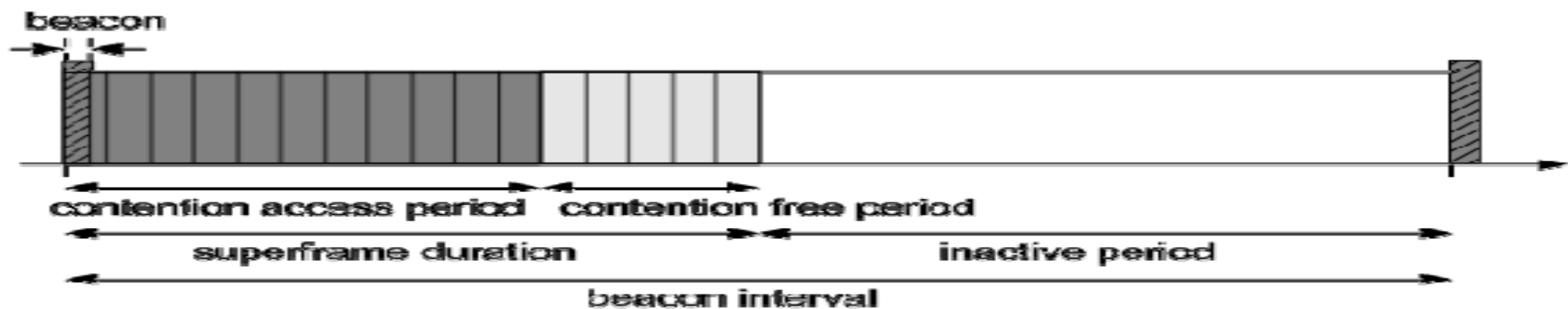
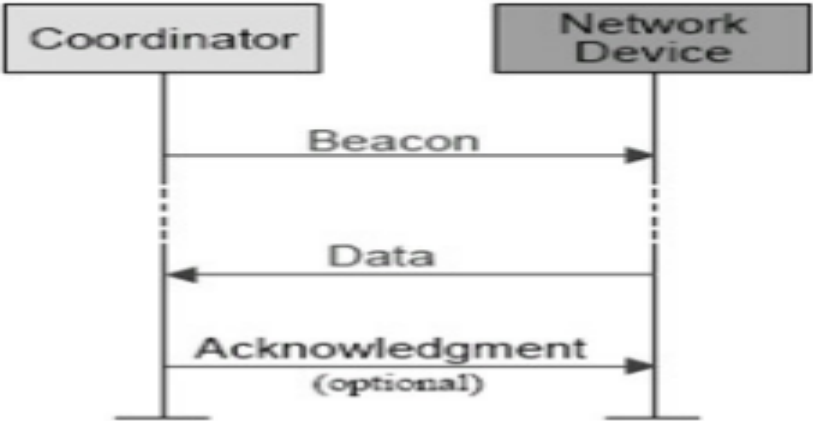
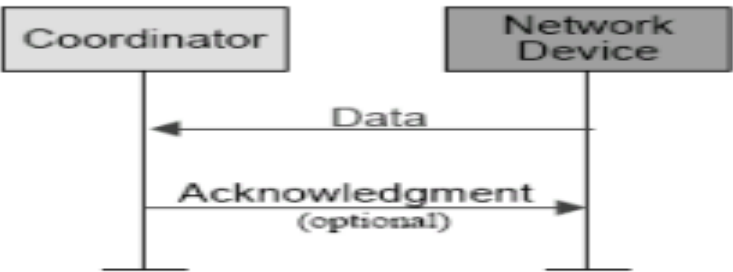


Figure 11: ZigBee Super Frame Structure

MAC Data Service Diagrams



ZigBee Advantages

- Easy to install and implement.
- It has low cost, low power, and long battery life.
- It supports many nodes (around 6500).
- It is more reliable and self-healing.

ZigBee Disadvantages

- It has a low data rate. {250kbps MAX}
- It is not as secure as Wi-Fi and Bluetooth.
- It requires additional devices ZCs and ZRs, which increases the cost.
- It lacks internet protocol support.
- ZigBee Networks are incompatible with other networks.